PRAKTIKUM 2

DATA DEFINITION LANGUAGE (DDL)

Nama : **Muhammad Slamet Ridho Novianto**

Kelas : **2 STr TRI B**

NRP : **2425600050**

A. Tujuan

1. Mengetahui dan memahami memahami SQL dan perintah DDL pada SQL.
2. Dapat membuat table dengan benar beserta relationshipnya.
3. Memahami type table InnoDB.

B. Soal Pendahuluan

1. Deklarasikan perintah SQL untuk:
    a. Membuat database
    b. Melihat seluruh database pada mysql server
    c. Mengakses database/ menggunakan database
    d. Menghapus database

Jawaban Soal Pendahuluan

1. Deklarasi perintah sql untuk :

a. Membuat Database

```
mysql> CREATE DATABASE vian_table;
Query OK, 1 row affected (0.02 sec)
```

b. Melihat seluruh database pada server mysql

```
mysql> SHOW DATABASES;
+--------------------+
| Database           |
+--------------------+
| esms_app           |
| information_schema |
| mysql              |
| performance_schema |
| prak_50            |
| sys                |
| ticketing_system   |
| vian_table         |
+--------------------+
8 rows in set (0.05 sec)
```

c. Mengakses database/ menggunakan database

```
mysql> USE vian_table;
Database changed
```

d. Menghapus database

```
mysql> DROP DATABASE vian_table;
Query OK, 0 rows affected (0.04 sec)
```

C. Langkah Percobaan

- Membuat Table
  Saya membuat tabel di MySQL server dengan nama “praktikum”.

```
mysql> create database praktikum;
Query OK, 1 row affected (0.05 sec)
```

  - Create table
    Saya memasukkan perintah “USE praktikum**;”** supaya semua perintah pembuatan tabel masuk ke database “praktikum” tidak salah tempat. Lalu Saya memasukkan perintah “CREATE TABLE Mahasiswa (NRP CHAR(11), Nama VARCHAR(40), Alamat VARCHAR(100));” untuk mendefinisikan kolom-kolom awal seperti NRP, Nama, dan Alamat sesuai kebutuhan data mahasiswa.

```
mysql> use praktikum;
Database changed
mysql> CREATE TABLE Mahasiswa (  NRP    CHAR(11),  Nama   VARCHAR(40),  Alamat VARCHAR(100));
Query OK, 0 rows affected (0.05 sec)
```

  - Melihat Table dan Struktur Table
    Saya menjalankan perintah “SHOW TABLES**;”** untuk memastikan tabel yang saya buat benar-benar sudah tercatat di database aktif.

```
mysql> show tables;
+---------------------+
| Tables_in_praktikum |
+---------------------+
| mahasiswa           |
+---------------------+
1 row in set (0.03 sec)
```

    Saya menjalankan perintah “DESC Mahasiswa;” untuk melihat detail kolom (nama kolom dan tipe datanya).

```
mysql> desc mahasiswa;
+--------+--------------+------+-----+---------+-------+
| Field  | Type         | Null | Key | Default | Extra |
+--------+--------------+------+-----+---------+-------+
| NRP    | char(11)     | YES  |     | NULL    |       |
| Nama   | varchar(40)  | YES  |     | NULL    |       |
| Alamat | varchar(100) | YES  |     | NULL    |       |
+--------+--------------+------+-----+---------+-------+
3 rows in set (0.01 sec)
```

- Constrain Table

  Constraint adalah aturan untuk menjaga **konsistensi** dan **integritas** data. Jenis yang umum dipakai meliputi primary key, foreign key, unique, not null, dan check.

  - Membuat database “order entry”
    Saya membuat database “OrderEntry” lalu “USE” database tersebut supaya semua tabel relasi dibuat dalam 1 tempat yang rapi.

```
mysql> CREATE DATABASE OrderEntry;USE OrderEntry;
Query OK, 1 row affected (0.01 sec)

Database changed
```

  - Membuat table inti dulu vendor dan customer
    Saya membuat tabel yang berdiri sendiri lebih dulu (contohnya vendors dan customers).

```
mysql> CREATE TABLE vendors (  vend_id      CHAR(4)      NOT NULL PRIMARY KEY,  vend_name
VARCHAR(25)  NOT NULL,  vend_address VARCHAR(30),  vend_city    VARCHAR(20),  vend_state   VA
RCHAR(5),  vend_zip     VARCHAR(7),  vend_country VARCHAR(15));
Query OK, 0 rows affected (0.07 sec)

mysql> CREATE TABLE customers (  cust_id      CHAR(5)      NOT NULL PRIMARY KEY,  cust_name
  VARCHAR(25)  NOT NULL,  cust_address VARCHAR(30)  NULL,  cust_city    VARCHAR(25)  NULL,  c
ust_state   VARCHAR(5)   NULL,  cust_zip     VARCHAR(5)   NULL,  cust_country VARCHAR(20)  NU
LL,  cust_contact VARCHAR(25)  NULL,  cust_email   VARCHAR(30)  NULL);
Query OK, 0 rows affected (0.07 sec)

mysql> show tables;
+----------------------+
| Tables_in_orderentry |
+----------------------+
| customers            |
| vendors              |
+----------------------+
2 rows in set (0.00 sec)
```

- Membuat table products dan table orders

  Saya membuat tabel products dan orders setelah tabel inti ada.

```
mysql> CREATE TABLE products (  prod_id    VARCHAR(10) NOT NULL PRIMARY KEY,  vend_id    CHAR
(4)     NOT NULL,  prod_name  VARCHAR(25) NOT NULL,  prod_price INT         NOT NULL,  prod_d
esc  VARCHAR(255) NULL);
Query OK, 0 rows affected (0.02 sec)

mysql> CREATE TABLE orders (  order_num  INT     NOT NULL,  order_date DATE    NOT NULL,  cus
t_id    CHAR(5) NOT NULL,  PRIMARY KEY (order_num));
Query OK, 0 rows affected (0.02 sec)

mysql> show tables;
+----------------------+
| Tables_in_orderentry |
+----------------------+
| customers            |
| orders               |
| products             |
| vendors              |
+----------------------+
4 rows in set (0.00 sec)
```

- Membuat table productnotes dan table orderItems

  Saya membuat tabel productnotes dan orderItems setelah tabel inti ada.

```
mysql> CREATE TABLE productnotes (  note_id   CHAR(3)     NOT NULL,  prod_id   VARCHAR(10) NO
T NULL,  note_date DATE        NOT NULL,  note_text VARCHAR(200) NULL,  PRIMARY KEY (note_id)
,  FOREIGN KEY (prod_id) REFERENCES products (prod_id));
Query OK, 0 rows affected (0.04 sec)

mysql> CREATE TABLE orderitems (  order_num  INT         NOT NULL,  order_item INT         NO
T NULL,  prod_id    VARCHAR(10) NOT NULL,  quantity   INT         NOT NULL,  PRIMARY KEY (ord
er_num, order_item));
Query OK, 0 rows affected (0.02 sec)

mysql> show tables;
+----------------------+
| Tables_in_orderentry |
+----------------------+
| customers            |
| orderitems           |
| orders               |
| productnotes         |
| products             |
| vendors              |
+----------------------+
6 rows in set (0.00 sec)
```

- Cek Hasil

```
mysql> SHOW TABLES;DESC vendors;DESC customers;DESC products;DESC orders;DESC productnotes;DESC orderitems;
+----------------------+
| Tables_in_orderentry |
+----------------------+
| customers            |
| orderitems           |
| orders               |
| productnotes         |
| products             |
| vendors              |
+----------------------+
6 rows in set (0.00 sec)

+--------------+-------------+------+-----+---------+-------+
| Field        | Type        | Null | Key | Default | Extra |
+--------------+-------------+------+-----+---------+-------+
| vend_id      | char(4)     | NO   | PRI | NULL    |       |
| vend_name    | varchar(25) | NO   |     | NULL    |       |
| vend_address | varchar(30) | YES  |     | NULL    |       |
| vend_city    | varchar(20) | YES  |     | NULL    |       |
| vend_state   | varchar(5)  | YES  |     | NULL    |       |
| vend_zip     | varchar(7)  | YES  |     | NULL    |       |
| vend_country | varchar(15) | YES  |     | NULL    |       |
+--------------+-------------+------+-----+---------+-------+
7 rows in set (0.00 sec)

+--------------+-------------+------+-----+---------+-------+
| Field        | Type        | Null | Key | Default | Extra |
+--------------+-------------+------+-----+---------+-------+
| cust_id      | char(5)     | NO   | PRI | NULL    |       |
| cust_name    | varchar(25) | NO   |     | NULL    |       |
| cust_address | varchar(30) | YES  |     | NULL    |       |
| cust_city    | varchar(25) | YES  |     | NULL    |       |
| cust_state   | varchar(5)  | YES  |     | NULL    |       |
| cust_zip     | varchar(5)  | YES  |     | NULL    |       |
| cust_country | varchar(20) | YES  |     | NULL    |       |
| cust_contact | varchar(25) | YES  |     | NULL    |       |
| cust_email   | varchar(30) | YES  |     | NULL    |       |
+--------------+-------------+------+-----+---------+-------+
9 rows in set (0.00 sec)

+------------+--------------+------+-----+---------+-------+
| Field      | Type         | Null | Key | Default | Extra |
+------------+--------------+------+-----+---------+-------+
| prod_id    | varchar(10)  | NO   | PRI | NULL    |       |
| vend_id    | char(4)      | NO   |     | NULL    |       |
| prod_name  | varchar(25)  | NO   |     | NULL    |       |
| prod_price | int          | NO   |     | NULL    |       |
| prod_desc  | varchar(255) | YES  |     | NULL    |       |
+------------+--------------+------+-----+---------+-------+
5 rows in set (0.00 sec)
```

```
+------------+---------+------+-----+---------+-------+
| Field      | Type    | Null | Key | Default | Extra |
+------------+---------+------+-----+---------+-------+
| order_num  | int     | NO   | PRI | NULL    |       |
| order_date | date    | NO   |     | NULL    |       |
| cust_id    | char(5) | NO   |     | NULL    |       |
+------------+---------+------+-----+---------+-------+
3 rows in set (0.00 sec)

+-----------+--------------+------+-----+---------+-------+
| Field     | Type         | Null | Key | Default | Extra |
+-----------+--------------+------+-----+---------+-------+
| note_id   | char(3)      | NO   | PRI | NULL    |       |
| prod_id   | varchar(10)  | NO   | MUL | NULL    |       |
| note_date | date         | NO   |     | NULL    |       |
| note_text | varchar(200) | YES  |     | NULL    |       |
+-----------+--------------+------+-----+---------+-------+
4 rows in set (0.00 sec)

+------------+-------------+------+-----+---------+-------+
| Field      | Type        | Null | Key | Default | Extra |
+------------+-------------+------+-----+---------+-------+
| order_num  | int         | NO   | PRI | NULL    |       |
| order_item | int         | NO   | PRI | NULL    |       |
| prod_id    | varchar(10) | NO   |     | NULL    |       |
| quantity   | int         | NO   |     | NULL    |       |
+------------+-------------+------+-----+---------+-------+
4 rows in set (0.00 sec)
```

- Nilai Otomatis dan Nilai Default
  - Nilai otomatis / Auto Increment

    Saya membuat tabel Mahasiswa2 dan menjadikan kolom ID sebagai primary key auto increment supaya nomor ID terisi otomatis oleh sistem.

```
mysql> USE praktikum;
Database changed
mysql>
mysql> CREATE TABLE Mahasiswa2 (
    ->   ID           INT(5) NOT NULL PRIMARY KEY AUTO_INCREMENT,
    ->   NRP          CHAR(8) NOT NULL,
    ->   Nama_Mhs     VARCHAR(50),
    ->   ProgramStudi VARCHAR(200),
    ->   Departemen   VARCHAR(30)
    -> );
Query OK, 0 rows affected, 1 warning (0.02 sec)

mysql>
mysql> SHOW TABLES;
+---------------------+
| Tables_in_praktikum |
+---------------------+
| mahasiswa           |
| mahasiswa2          |
| mtkul               |
+---------------------+
3 rows in set (0.00 sec)

mysql> DESC Mahasiswa2;
+--------------+--------------+------+-----+---------+----------------+
| Field        | Type         | Null | Key | Default | Extra          |
+--------------+--------------+------+-----+---------+----------------+
| ID           | int          | NO   | PRI | NULL    | auto_increment |
| NRP          | char(8)      | NO   |     | NULL    |                |
| Nama_Mhs     | varchar(50)  | YES  |     | NULL    |                |
| ProgramStudi | varchar(200) | YES  |     | NULL    |                |
| Departemen   | varchar(30)  | YES  |     | NULL    |                |
+--------------+--------------+------+-----+---------+----------------+
5 rows in set (0.00 sec)
```

- Nilai Default

  Saya menampilkan tabel dan struktur tabel untuk memastikan kolom ID benar-benar sudah jadi Priary Key dan auto increment.

```
mysql> USE praktikum;
Database changed
mysql>
mysql> CREATE TABLE Mtkul (
    ->   Kode_Mtkul INT(5) NOT NULL PRIMARY KEY,
    ->   Nama_Mtkul VARCHAR(30),
    ->   Sks        INT(1) DEFAULT 0,
    ->   Semester   INT(1) DEFAULT 0
    -> );
Query OK, 0 rows affected, 3 warnings (0.03 sec)

mysql>
mysql> DESC Mtkul;
+------------+-------------+------+-----+---------+-------+
| Field      | Type        | Null | Key | Default | Extra |
+------------+-------------+------+-----+---------+-------+
| Kode_Mtkul | int         | NO   | PRI | NULL    |       |
| Nama_Mtkul | varchar(30) | YES  |     | NULL    |       |
| Sks        | int         | YES  |     | 0       |       |
| Semester   | int         | YES  |     | 0       |       |
+------------+-------------+------+-----+---------+-------+
4 rows in set (0.00 sec)
```

- Menentukan mesin penyimpanan tabel (ENGINE InnoDB)
  - Saya memakai InnoDB karena menurut saya lebih cocok untuk tabel yang berelasi (ada foreign key).

```
mysql> CREATE TABLE compress (
    ->   x INT NOT NULL PRIMARY KEY,
    ->   y VARCHAR(50) NULL
    -> ) ENGINE = InnoDB;
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> SHOW TABLES;
+---------------------+
| Tables_in_praktikum |
+---------------------+
| compress            |
| mahasiswa           |
| mahasiswa2          |
| mtkul               |
+---------------------+
4 rows in set (0.00 sec)

mysql> DESC compress;
+-------+-------------+------+-----+---------+-------+
| Field | Type        | Null | Key | Default | Extra |
+-------+-------------+------+-----+---------+-------+
| x     | int         | NO   | PRI | NULL    |       |
| y     | varchar(50) | YES  |     | NULL    |       |
+-------+-------------+------+-----+---------+-------+
2 rows in set (0.00 sec)
```

D. Tugas

1. Buatlah sebuah database dengan nama “Penjualan_Barang”
2. Dengan menggunakan database penjualan_barang buatlah tabel hasil dari diagram relationship penjualan barang.
3. Tampilkan hasil capture dari command SQL beserta hasilnya.

Jawaban Tugas

1. Saya membuat database dengan nama “penjualan_barang”

```
mysql> CREATE DATABASE penjualan_barang;USE penjualan_barang;
Query OK, 1 row affected (0.01 sec)

Database changed
```

2. Membuat table hasil table diagram relationship.

```
mysql> CREATE TABLE pelanggan (
    ->   id_pelanggan INT NOT NULL AUTO_INCREMENT,
    ->   nama VARCHAR(50) NOT NULL,
    ->   PRIMARY KEY (id_pelanggan)
    -> );
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> CREATE TABLE barang (
    ->   id_barang INT NOT NULL AUTO_INCREMENT,
    ->   nama_barang VARCHAR(50) NOT NULL,
    ->   harga DECIMAL(12,2) NOT NULL,
    ->   PRIMARY KEY (id_barang)
    -> );
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> CREATE TABLE nota_penjualan (
    ->   id_nota INT NOT NULL AUTO_INCREMENT,
    ->   tgl_nota DATE NOT NULL,
    ->   id_pelanggan INT NOT NULL,
    ->   PRIMARY KEY (id_nota),
    ->   FOREIGN KEY (id_pelanggan) REFERENCES pelanggan(id_pelanggan)
    -> );
Query OK, 0 rows affected (0.04 sec)

mysql>
mysql> CREATE TABLE detail_jual (
    ->   id_nota INT NOT NULL,
    ->   id_barang INT NOT NULL,
    ->   jumlah INT NOT NULL,
    ->   PRIMARY KEY (id_nota, id_barang),
    ->   FOREIGN KEY (id_nota) REFERENCES nota_penjualan(id_nota),
    ->   FOREIGN KEY (id_barang) REFERENCES barang(id_barang)
    -> );
Query OK, 0 rows affected (0.04 sec)
```

3. Hasil capture command sql serta hasil commandnya.

```
mysql> SHOW TABLES;DESC pelanggan;DESC barang;DESC nota_penjualan;DESC detail_jual;
+-----------------------------+
| Tables_in_penjualan_barang |
+-----------------------------+
| barang                      |
| detail_jual                 |
| nota_penjualan              |
| pelanggan                   |
+-----------------------------+
4 rows in set (0.00 sec)

+---------------+-------------+------+-----+---------+----------------+
| Field         | Type        | Null | Key | Default | Extra          |
+---------------+-------------+------+-----+---------+----------------+
| id_pelanggan  | int         | NO   | PRI | NULL    | auto_increment |
| nama          | varchar(50) | NO   |     | NULL    |                |
+---------------+-------------+------+-----+---------+----------------+
2 rows in set (0.00 sec)

+-------------+---------------+------+-----+---------+----------------+
| Field       | Type          | Null | Key | Default | Extra          |
+-------------+---------------+------+-----+---------+----------------+
| id_barang   | int           | NO   | PRI | NULL    | auto_increment |
| nama_barang | varchar(50)   | NO   |     | NULL    |                |
| harga       | decimal(12,2) | NO   |     | NULL    |                |
+-------------+---------------+------+-----+---------+----------------+
3 rows in set (0.00 sec)

+---------------+------+------+-----+---------+----------------+
| Field         | Type | Null | Key | Default | Extra          |
+---------------+------+------+-----+---------+----------------+
| id_nota       | int  | NO   | PRI | NULL    | auto_increment |
| tgl_nota      | date | NO   |     | NULL    |                |
| id_pelanggan  | int  | NO   | MUL | NULL    |                |
+---------------+------+------+-----+---------+----------------+
3 rows in set (0.00 sec)

+-----------+------+------+-----+---------+-------+
| Field     | Type | Null | Key | Default | Extra |
+-----------+------+------+-----+---------+-------+
| id_nota   | int  | NO   | PRI | NULL    |       |
| id_barang | int  | NO   | PRI | NULL    |       |
| jumlah    | int  | NO   |     | NULL    |       |
+-----------+------+------+-----+---------+-------+
3 rows in set (0.00 sec)
```